

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	12 July 2026
SID	SWUID2026198226
Project Name	AI-Powered Customer Support & ServiceAutomation.
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example:

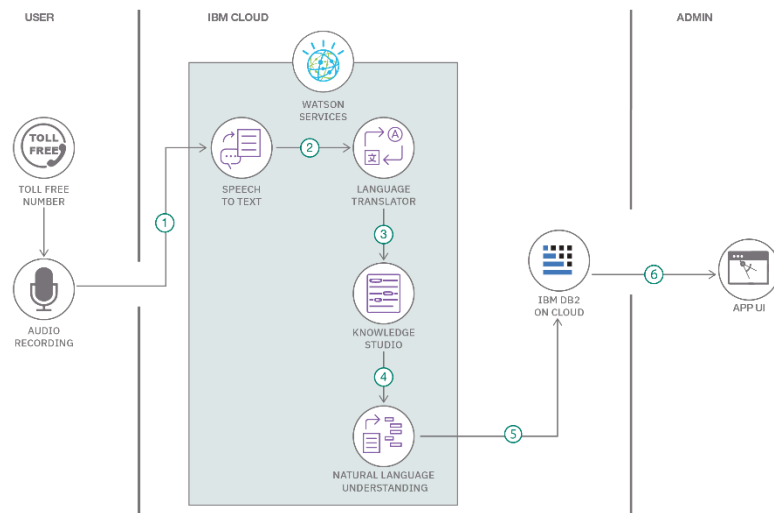


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Multi-channel interaction: Lightning Console for Support Agents to manage case feeds, and integrated Salesforce Experience Cloud or Mobile SDK for end-customers to submit cases.	Salesforce Lightning Web Components (LWC), Experience Cloud
2.	Application Logic-1	Core orchestration logic: Triage process, record evaluation, and executing workflow status changes on custom objects.	Salesforce Record-Triggered Flows & Apex (Trigger Frameworks)
3.	Application Logic-2	Real-time text & payload ingestion: Ingesting unstructured customer support data streams and mapping variables into CRM fields.	Salesforce Service Cloud Ingestion Engine / REST API
4.	Application Logic-3	Conversational Interface: Autonomous AI agent handling initial customer triage, natural language interactions, and automated order status queries.	Agentforce (Salesforce Foundations)
5.	Database	Transactional Data: Storing tightly structured CRM data including relational profiles, detailed order histories, and internal staff records.	Salesforce Standard & Custom Objects (Customer, Merchandise, Product, Order)
6.	Cloud Database	Dynamic Case Logs: Storing high-volume interaction logs, real-time chat histories, and unstructured support tickets securely within the CRM data layer.	Salesforce Data Cloud / User Case Custom Object
7.	File Storage	Static Assets: Secure, cloud-native storage for case-related attachments, damaged merchandise photos, and archived text transcripts.	Salesforce Files (ContentVersion Architecture)
8.	External API-1	Context Enrichment: Used to extract and pass rich CRM metadata context into AI execution contexts for complex order triage.	Salesforce Prompt Builder (MetadataInspection template)
9.	External API-2	Operational Integration: External service execution triggered from flows to process authorized order refunds and billing adjustments.	Salesforce External Services / 3rd Party Payment Gateway API

10.	Machine Learning Model	Automated Sentiment Analysis & Summaries: Running real-time sentiment analysis and auto-generating structured case summaries securely.	Salesforce Einstein Trust Layer / Large Language Models (LLMs) via Prompt Builder
11.	Infrastructure (Server / Cloud)	Cloud Deployment: Enterprise multi-tenant cloud infrastructure natively executing the entire customer service automation architecture.	Salesforce Core Cloud Infrastructure (Lightning Platform)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Multi-Tenant Cloud Architecture	Replaces traditional open-source staging by deploying the UI (LWC) and backend logic natively on the secure, shared Salesforce Lightning platform.	Salesforce LWC (Frontend) & Lightning Platform (Backend)
2.	Security & Trust Implementations	Provides end-to-end data security and compliance via Shield encryption, robust OAuth 2.0 access controls, and strict data masking through the Einstein Trust Layer.	Salesforce Einstein Trust Layer, Shield Platform Encryption (AES-256), OAuth 2.0
3.	Native Elastic Scalability	Automates resource allocation to handle large volumes of unstructured case logs and transaction spikes without manual infrastructure configuration.	Multi-Tenant Resource Allocation & Salesforce Data Cloud
4.	Availability	Relies on global Salesforce data centers to ensure persistent uptime, geographic redundancy, and immediate failover capabilities.	Salesforce Global Infrastructure Redundancy (Multi-AZ)
5.	Performance	Employs optimized flow execution engines, native platform caching, and built-in CDNs to keep case routing and AI processing times under 3 seconds.	Record-Triggered Flow Engine, Platform Cache, and native Salesforce CDN

